

UNIT - A

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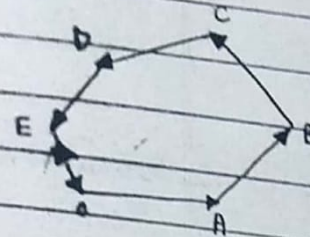
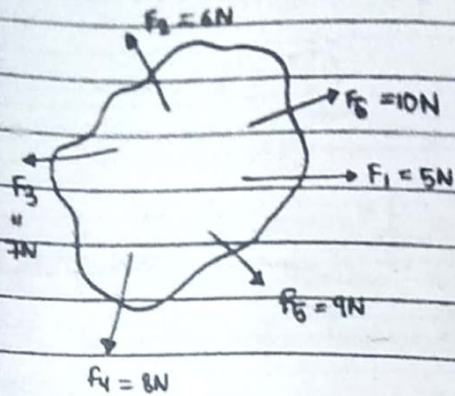
COPLANAR FORCES:

1) Concurrent

2) Parallel

for Concurrent Forces:

Polygon Law of forces:
let us assume, $1\text{cm} = 1\text{N}$



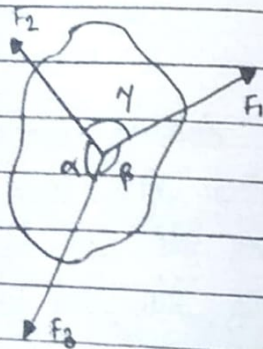
for equilibrium: $\sum F_x = 0$
 $\sum F_y = 0$

$\sum \text{Moment} = 0$

and $R = \sqrt{\sum F_x^2 + \sum F_y^2}$

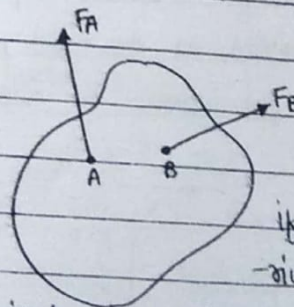
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Lamme's Theorem:



$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

• for a two force body,



if the body is in equilibrium then it is

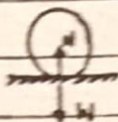
necessary that the magnitude and direction should be same and opposite respectively and there line of action should be collinear.

$$F_A = F_B$$

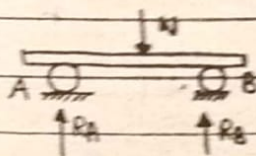
- For equilibrium of body under the action of 3 forces:
 - 1) Resultant of 2 forces should be equal to 3rd,
 - 2) Resultant = F_3
 - 3) Line of action of all 3 should meet at common point.

Type of Supports

- 1) Frictionless Support:

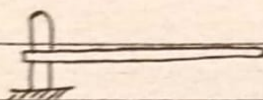


- 2) Roller Support:



It only checks the vertical movement.

- 3) Hinged Support:

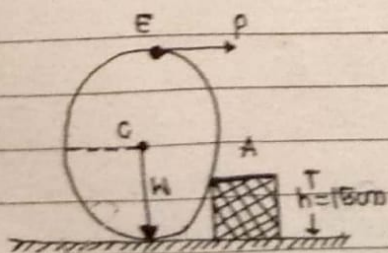


It checks the vertical and horizontal movement but allows the to and fro movement.

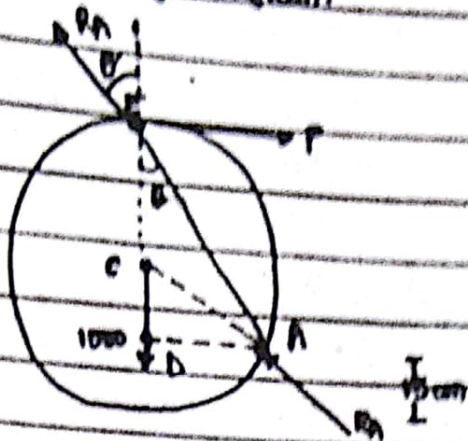
- 4) Fixed support:

Example

- Q. A uniform wheel of 60cm diameter and weighing 1000N rests against a rectangular block 15cm high lying on a horizontal plane. It is to be pulled over this block by a horizontal force P applied to the end of a string. Find the force P when the wheel is just about to roll over the block.



Free body Diagram:



Moment about A:

$$\sum M_A = 0$$

Moment of RA is zero.

$$P(45) + W(25.9)(-1) = 0$$

$$P(45) - 25.9(1000)$$

$$P = \frac{25.9 \times 1000}{45}$$

$$P = 575.55 \text{ N}$$

$$AD = 15 \tan \theta$$

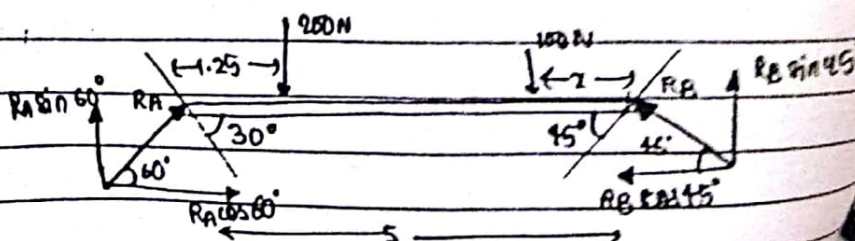
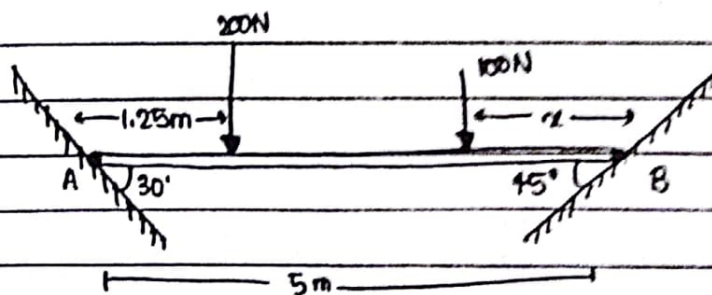
$$CA^2 = CB^2 + AD^2$$

$$(20)^2 = (15)^2 + AD^2$$

$$AD^2 = 645$$

$$AD = 25.9$$

A weightless bar AB is placed in horizontal position on smooth inclined as shown in the figure. Calculate the distance x at which the 100N load may be placed from end B so that the bar remains horizontal.



$$\sum F_x = 0$$

$$R_A \cos 60^\circ = R_B \cos 45^\circ$$

$$R_A \cdot \frac{1}{2} = \frac{R_B}{\sqrt{2}}$$

$$R_A = \sqrt{2} R_B$$

$$R_B = 155.86$$

$$\sum F_y = 0$$

$$R_A \sin 60^\circ + R_B \sin 45^\circ = 200$$

$$R_A \frac{\sqrt{3}}{2} + \frac{R_B}{\sqrt{2}} = 200$$

$$\Rightarrow \frac{\sqrt{3}}{2} R_A + \frac{R_B}{\sqrt{2}} = 200$$

$$\Rightarrow \frac{\sqrt{3}}{2} R_A + R_B = 200\sqrt{2}$$

$$\Rightarrow R_B = \frac{200\sqrt{2}}{\sqrt{3}+1}$$

$$R_A = 219.65$$

moment about B:

$$100(x) + 200(3.75) + R_A \sin 60^\circ \cdot 5(1) = 0$$

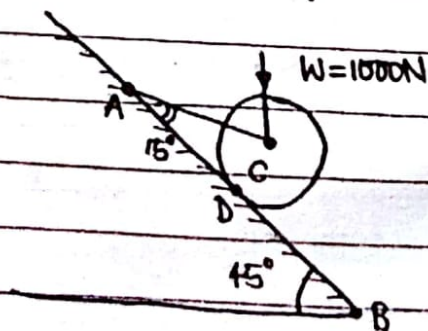
$$100x + 750 = R_A \cdot \frac{\sqrt{3}}{2} \times 5$$

$$100x + 750 = 219.65 \times \frac{\sqrt{3}}{2} \times 5$$

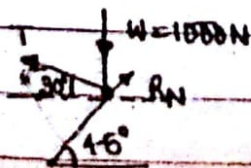
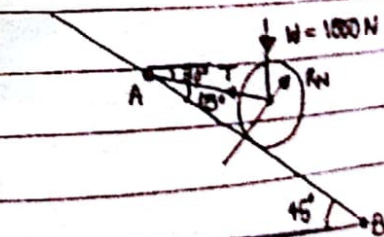
$$x = 2.01 \text{ m}$$

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Q. A roller of weight, $W = 1000 \text{ N}$ rests on a smooth inclined plane. It is kept away from rolling down the plane by a string. Find the tension in the string and reaction at the point of contact D.



Sol.ⁿ



$$\sum F_x = 0$$

$$T \times \frac{\sqrt{3}}{2} = R_N \times \frac{1}{\sqrt{2}} \quad \text{--- (1)}$$

$$T \times \frac{1}{2} + R_N \times \frac{1}{\sqrt{2}} = 1000 \quad \text{--- (2)}$$

Using eq. (1) and eq. (2): $\frac{T}{2} + \frac{T\sqrt{3}}{2} = 1000$

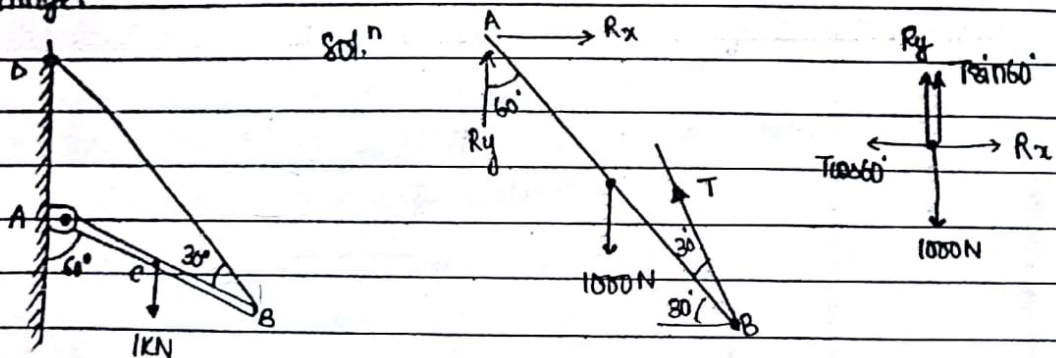
$$T(1+\sqrt{3}) = 2000$$

$$T = \frac{2000}{2.73} = 732.6 \text{ N}$$

$$\frac{T \times \sqrt{3}}{2} = \frac{R_N}{\sqrt{2}}$$

$$R_N = \frac{\sqrt{6}}{2} \times 732 = 897 \text{ N}$$

Q. A bar AB of weight 1 kN is hinged to a vertical wall at A and supported at the end by a cable BD. Find the tension in the cable and magnitude and direction of reaction at the hinge.



$$R_x = T \cos 60^\circ$$

$$R_y + T \sin 60^\circ = 1000$$

$$\sum \text{Moment at A} = 0$$

$$AB = l$$

$$\cos 30^\circ \times 1000 \left(\frac{l}{2}\right) + T \cos 60^\circ \cdot l \sin 30^\circ = T \sin 60^\circ \times l \cos 30^\circ$$

$$\frac{\sqrt{3}}{4} \times 1000 + \frac{T}{4} = \frac{3T}{4}$$

$$1000 \sqrt{3} = 2T$$

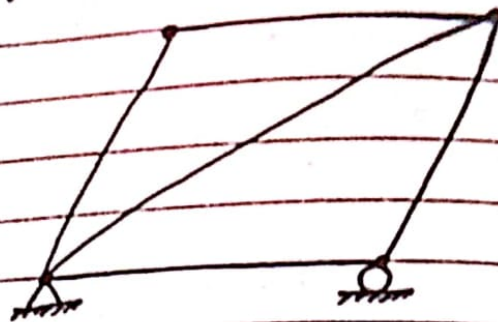
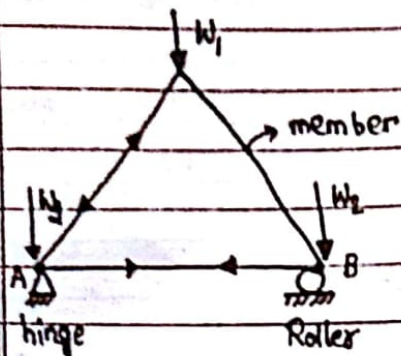
$$500 \sqrt{3} = T = 866 \text{ N}$$

$$R_x = 433$$

$$R_y = 1000 - 750 = 250$$

$$R_A = \sqrt{(433)^2 + (250)^2} = \sqrt{187489 + 62500} = \sqrt{249989} = 500$$

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Analysis of Truss

4-Joint and 5 Member

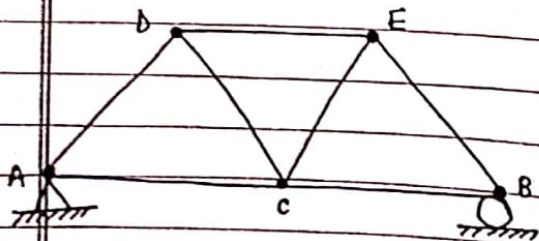
Pull - Tension

Push - Compression

Perfect Truss Condition: No. of members in a joint = m
 No. of joints = $n = j$

For equilibrium: $m = 2j - 3$

Load is always applied ^{won't} on the joint not on any member of the truss. Otherwise it forms a Two-force Member.



No. of joints = 5

No. of Members = 7

It's a perfect truss.

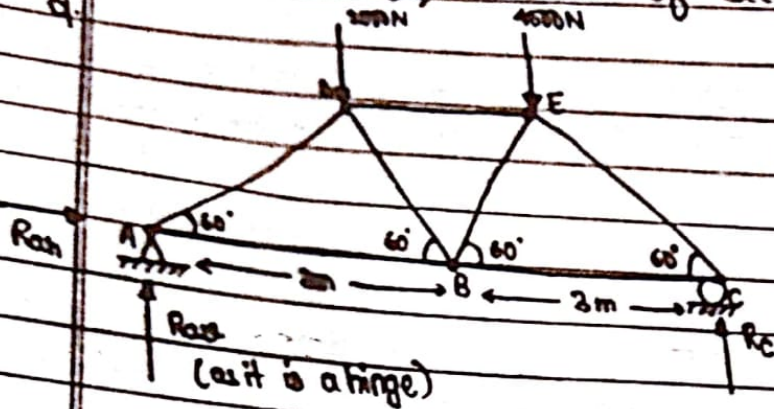
Assumptions of Truss:

- 1) Joints are frictionless.
- 2) Load is applied on the joints only.
- 3) Each member of a truss is a two force member.
- 4) A perfect truss is called statically determinate.
- 5) Weight of the member is neglected.

Three methods to find internal stress in a truss:

- 1) Method of Joints
- 2) Method of Section
- 3) Graphical Method. (not in syllabus)

1) Method of Section:



$$\sum F_x = 0$$

(Internal forces are not considered)

$$R_{AH} = 0$$

(no force in x-direction)

$$\sum \text{Moment at A} = 0$$

$$R_C \times 6 = 2500 \times 1.5 + 4500 \times 4.5$$

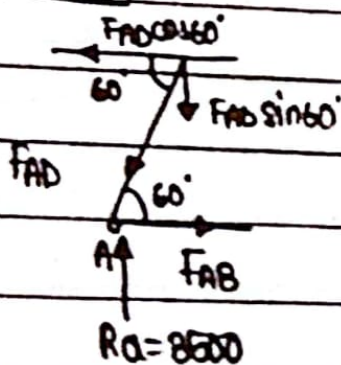
$$R_C = \frac{3000}{6} + \frac{18000}{6}$$

$$R_C = \frac{21000}{6} = 3500 \text{ N}$$

$$\text{and } R_A = 2500 \text{ N}$$

Method of Joints:

Joint A:



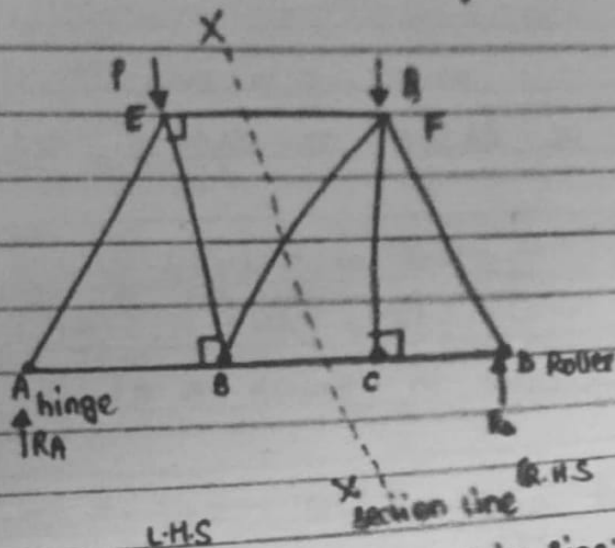
$$F_{AB} - F_{AD} \times \frac{1}{2} = 0$$

$$A \rightarrow C \rightarrow B$$

$$F_{AB} \times \frac{\sqrt{3}}{2} = 3500 = R_C$$

- Always, move to the other point which does not contain more than 2 forces. Thus we will consider C after joint.

2) Method of Sections



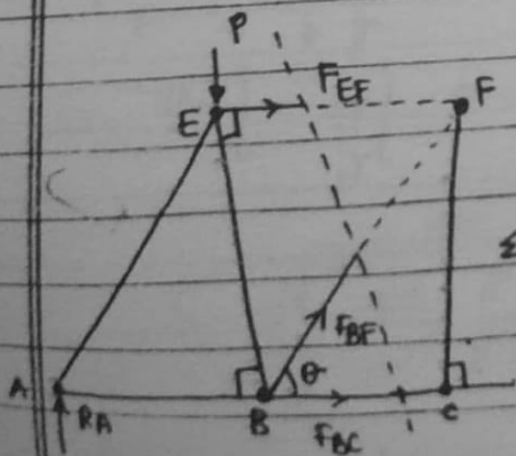
Find the force in member EF?

By drawing section line, we divide truss in two equal parts.

Assumptions for drawing section line:

- 1) Section line should pass through joints and should cut the member on which force is to be calculated.
- 2) It should divide the truss in two equal parts. (separate parts) but equality is not important.
- 3) It should be drawn in such a way that it should not cut more than 3 members at a time.
- 4) While considering the equilibrium of any side of truss then we can calculate the moment on any place (i.e. on L.H.S and R.H.S any of the side).

Equilibrium Diagram on L.H.S: Here External forces are P and RA. These the forces which were initially internal forces are now after drawing section line have become external forces. i.e. F_{EF} , F_{BF} , F_{BC} .



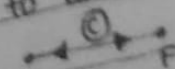
$$\sum (\text{Moment at B}) = 0$$

line of action of P is passing through B. Thus, moment of P, F_{BF} and F_{BC} is zero.

$$R_A \times AB + F_{EF} \times EB = 0$$

Here, F_{EF} comes out to be -ve.

Thus, force on member EF is compressive.



For calculating Force on member BC; we have to calculate the moment at E.

$$\sum (\text{Moment of } F) = 0$$

$$P \times EF + F_{BC} \times BE = R_A \times AC$$

$$F_{BC} = \frac{(R_A)(AC) - (P)(EF)}{(BE)}$$

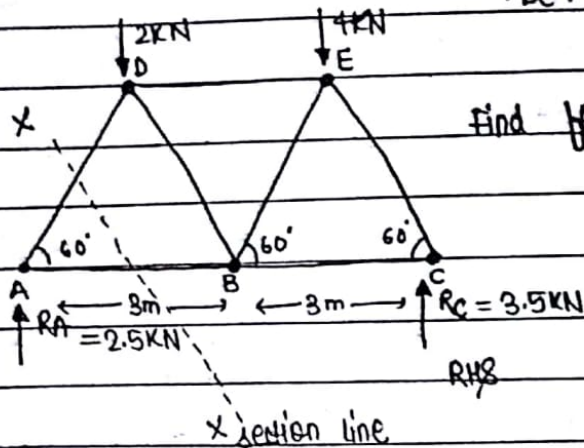
OR

For the whole truss: $\sum F_x = 0$

$$\sum F_y = 0$$

$$F_{BF} \sin \theta + R_A = P$$

$$F_{BC} + F_{EF} + F_{BF} \cos \theta = 0$$



find force in member AD, AB and BE.

LHS

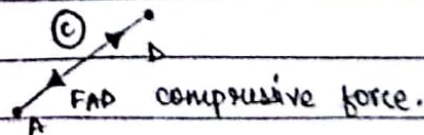
$$\sum (\text{Moment at B}) = 0$$

$$2.5(3) + F_{AD} \cdot \frac{\sqrt{3}}{2} \cdot 3 = 0$$

$$2.5(3000) + F_{AD} \cdot \frac{3\sqrt{3}}{2} = 0$$

$$(2500)3 = -F_{AD} \cdot \frac{\sqrt{3}}{2}(3)$$

$$F_{AD} = -\frac{5000}{\sqrt{3}} = -2.8 \text{ kN} \Rightarrow F_{AD}$$



FAD compressive force.

$$\sum (\text{Moment at D}) = 0$$

$$AD = 1.5 \tan 60^\circ = \frac{3\sqrt{3}}{2}$$

$$R_A(1.5) = F_{AB} \left(\frac{3\sqrt{3}}{2} \right)$$

$$F_{AB} = \frac{(2.5)(3)}{3\sqrt{3}} = \frac{2.5}{\sqrt{3}} = 1.44 \text{ kN}$$

Power transmitted by the belt $\Rightarrow P = (T_1 - T_2) \times V$
 Velocity of belt

- Purpose is to transfer max. power which means $(T_1 - T_2)$ should be maximum and which will be max. when frictional force is maximum.

AS $T_1 = T_2 + F$

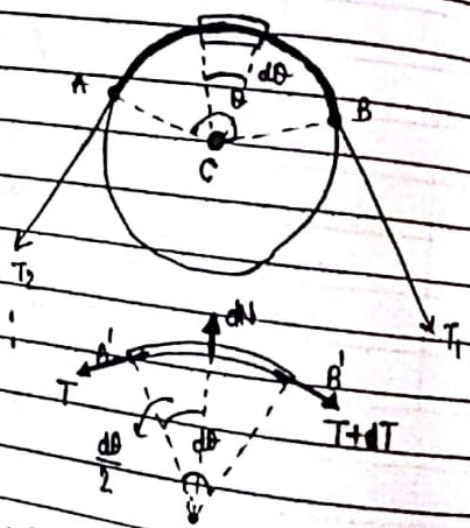
* $T_1 - T_2 = F$

Angle of lap: $\theta \propto$ Power transfer.



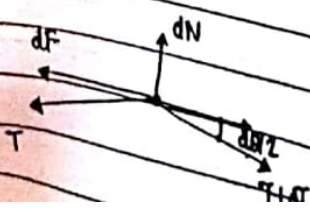
- $\rightarrow$ Flat belt: $\theta < 180$
- $\rightarrow$ Open belt: $\theta > 180$

- $T_1 > T_2 \rightarrow$ Clockwise
- $T_1 < T_2 \rightarrow$ Anti-clockwise



consider the eq.^m of belt only:

- Assume that it is moving clockwise.
- ext. of belt is very small as compared to pulley. so it is neglected.
- As belt is moving clockwise then frictional force will act in anti-clockwise dir.ⁿ.



$$\sum F_x = 0 \quad \frac{(T+dT)\cos\frac{d\theta}{2}}{2} = \frac{T\cos\frac{d\theta}{2}}{2} + \mu dN \quad - (1)$$

where μ is coefficient of friction.
 as $d\theta \ll 0$ so $\cos\frac{d\theta}{2} = 1$ and $\sin\frac{d\theta}{2} = \frac{d\theta}{2}$

$$T+dT = T + \mu dN$$

$$\boxed{dT = \mu dN} \quad - (2)$$

Increment in tension is dT .

$$\sum F_y = 0 \quad \frac{(T+dT)\sin\frac{d\theta}{2}}{2} = -\frac{T\sin\frac{d\theta}{2}}{2} + dN$$

$$(T+dT+T)\frac{d\theta}{2} = dN$$

$$2T + dT = \frac{2dN}{d\theta}$$

as $d\theta$ and dT both are too small
 $\Rightarrow \frac{2T + dT \cdot d\theta}{2} = dN$

$$\Rightarrow \boxed{\frac{dT}{T} = \mu d\theta} \quad - (3)$$

$$\boxed{dT = \mu T \cdot d\theta}$$

$$\frac{dT}{T} = \mu d\theta \rightarrow \text{for a small part}$$

$$\text{for the whole belt} \Rightarrow \int_{T_2}^{T_1} \frac{dT}{T} = \int_0^{\theta} \mu d\theta$$

$$\boxed{\frac{T_1}{T_2} = e^{\mu\theta}}$$

For $(T_1 - T_2)$ to be large:

- μ should be large which will increase value of T_1 .
- θ should be large

- Cross belt transfer more power as their angle of lap is 180° .

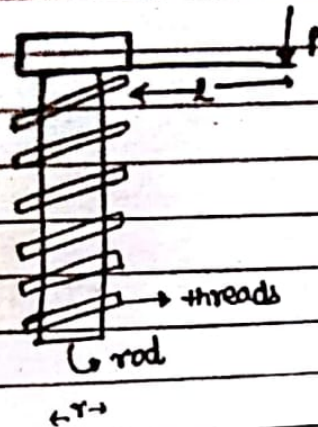
2) Screw Friction:

Screw jack is a self-locking machine.

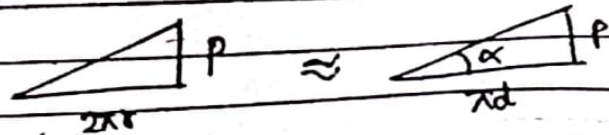
Levers is used:

load to be lifted with less effort.

$$\text{Mechanical Advantage} = \frac{W}{P}$$



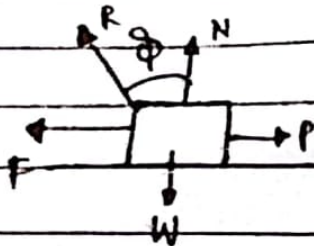
Threads are arranged in helical shape.



Gap b/w. two consecutive threads pitch, P .

$$\tan \alpha = \frac{P}{\pi d} \quad \text{where } \alpha \text{ is helix angle.}$$

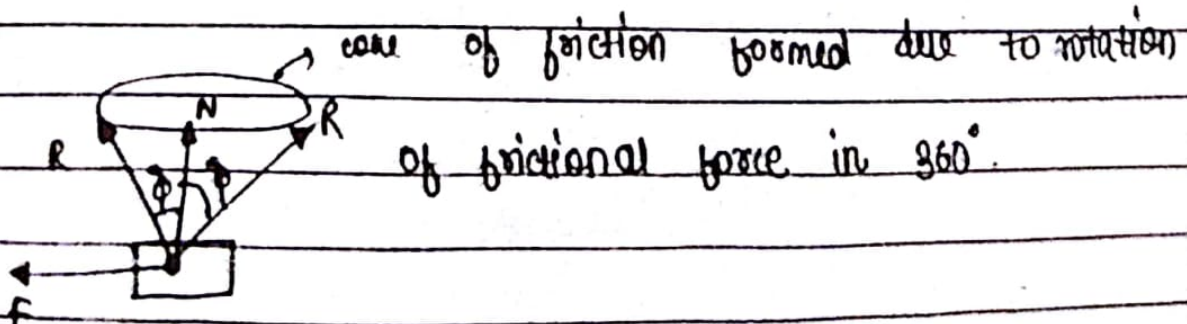
(# Angled Friction)



(ϕ) angle of friction depends on material of contact surfaces.

$$\mu = \tan \phi$$

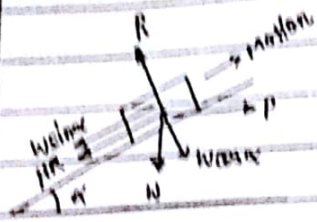
Angle of repose = Angle of friction.



Efficiency = $\frac{\text{Mechanical Advantage}}{\text{Velocity Ratio}} = \frac{M.A.}{V.R.}$

$V.R. = \frac{2\pi R}{\pi r}$

$\eta = \frac{\text{Ideal Effort}}{\text{Actual Effort}}$



$\sum F_x = 0$ and $\sum F_y = 0$

$P \cos \alpha = \mu R + W \sin \alpha$ — (1)

$R = P \sin \alpha + W \cos \alpha$ — (2)

$R \cos \alpha = \mu (P \sin \alpha + W \cos \alpha) + W \sin \alpha$

$P \cos \alpha = \mu P \sin \alpha + \mu W \cos \alpha + W \sin \alpha$

$P \cos \alpha - \mu P \sin \alpha = W (\mu \cos \alpha + \sin \alpha)$

$P = \frac{W (\mu \cos \alpha + \sin \alpha)}{(\cos \alpha - \mu \sin \alpha)}$

$\mu = \tan \phi = \frac{\sin \phi}{\cos \phi}$

$P = W \cdot \left(\frac{\sin \phi \cos \alpha}{\cos \phi} + \sin \alpha \right) = \frac{W \sin (\alpha + \phi)}{\cos (\alpha + \phi)}$

$\left(\cos \alpha - \frac{\sin \phi \cdot \sin \alpha}{\cos \phi} \right)$

$P = W \tan (\alpha + \phi)$

$\alpha \rightarrow$ helix angle

$\phi \rightarrow$ friction angle

$\eta = \frac{\text{Ideal Effort}}{\text{Actual Effort}} = \frac{W \tan \alpha}{W \tan (\alpha + \phi)}$

Irreversible Machine: The load is at its place even if we remove the effort. ($P=0$).

The helix angle for which η is maximum:-

$\frac{d\eta}{d\alpha} = 0$

$\therefore \alpha = \frac{\pi}{4} - \frac{\phi}{2}$

Q. A square thread of 7.5 cm mean diameter, is used to lift a wt. of 600N. Determine:

- (a) The tangential force which needs to be applied at the end of a lever of length 40 cm.
 (b) The efficiency of the screw jack and state whether the screw jack is self-locking or not. $P = 1.5$ cm.

$$P' \times l = P \times r$$

$$\mu = \tan \phi \Rightarrow \mu = 0.5$$

$$\tan \phi = 0.05$$

$$\phi = 2.86^\circ$$

$$\tan \alpha = \frac{P}{\pi d} = \frac{1.5}{3.14 \times 7.5}$$

$$\tan \alpha = 0.0636$$

$$\alpha = 3.63$$

$$P = 600 \tan(2.86 + 3.63)$$

$$P = 600 \tan(6.49)$$

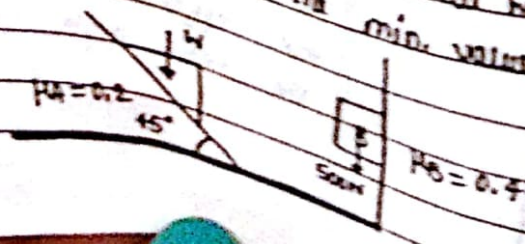
$$P \Rightarrow 68.2 \text{ N}$$

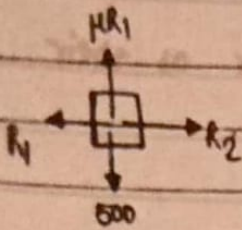
$$P' \times 40 = \frac{68.2 \times 7.5}{2}$$

$$P' = \frac{125.88 \times 9.5}{80} = 6.39 \text{ N}$$

$$\eta = \frac{\tan(3.63)}{\tan(6.49)} \times 100 = 56.7 \%$$

Q. A block of wt. W is placed on rough inclined plane having $\mu_A = 0.2$ and is held in position by means of a horizontal rod hinged to the block B which presses against a rough vertical wall having $\mu_B = 0.4$. The weight of block B is 500N. Find the min. value of W consistent with the equilibrium.





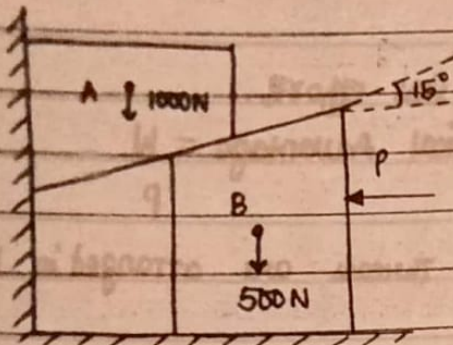
$$R_1 = R_2$$

$$\mu R_1 = 500$$

$$R_1 = \frac{500}{0.4} = 1250$$

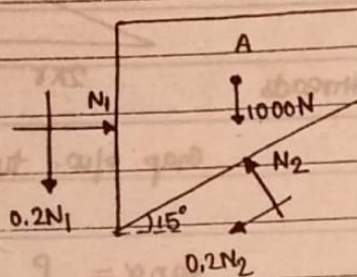
Wedge Friction:

Q. A Block of wt. 1000N is to be raised by means of a 15° wedge B weighing 500N. Assuming the coefficient of friction for all the contacts to be 0.2. Determine the min. force P that should be applied to raise the block.



$$\sum F_x = 0$$

$$\sum F_y = 0$$



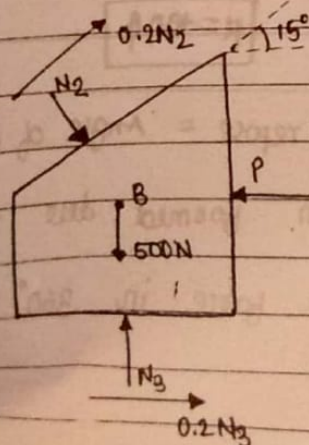
$$\sum F_x = 0$$

$$N_1 - 0.2N_2 \cos 15^\circ - N_2 \sin 15^\circ = 0$$

$$N_1 = (0.4522) N_2$$

$$\sum F_y = 0$$

$$0.2N_1 + 1000 + 0.2N_2 \sin 15^\circ = N_2 \cos 15^\circ \Rightarrow N_2 = 1214 \text{ N} \text{ \& } N_1 = 549 \text{ N}$$



$$\sum F_x = 0$$

$$0.2N_2 \cos 15^\circ + N_2 \sin 15^\circ + 0.2N_3 = P \quad \text{--- (1)}$$

$$N_2 \cos 15^\circ + 500 = 0.2N_2 \sin 15^\circ + N_3 \quad \text{--- (2)}$$

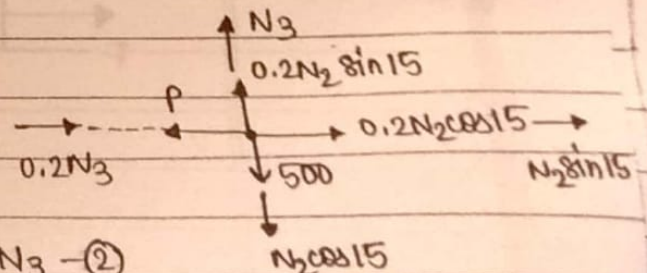
$$0.19 N_2 + 0.25 N_2 + 0.2 N_3 = P$$

$$0.44 N_2 - 0.05 N_2 - N_3 = -500$$

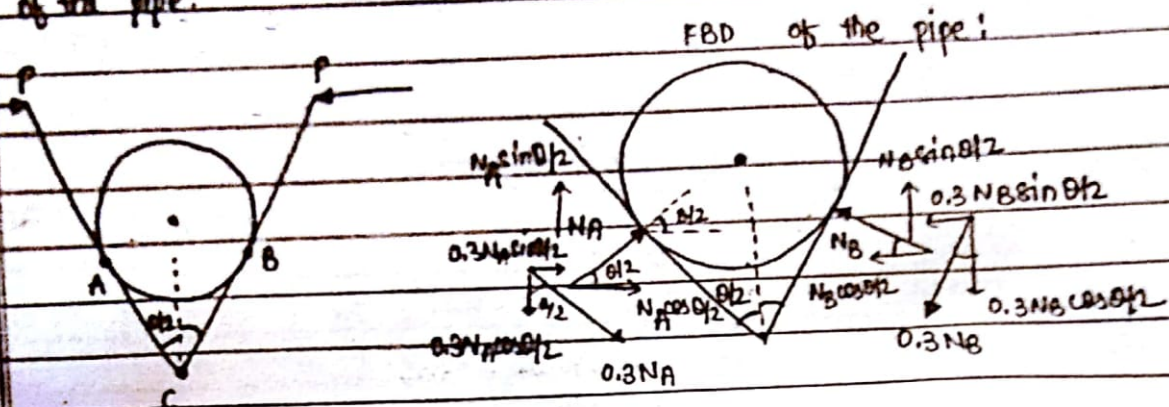
$$534.16 + 0.2 N_3 = P \quad \text{--- (3)}$$

$$N_3 = 1609.8 \text{ N}$$

$$P = 871 \text{ N}$$



A pipe is gripped by 2 levers that are joined at 'C'. If the coeff. of friction b/w the levers & the pipe is 0.3. Determine the angle θ . At which the pipe can be gripped with slip and neglect the wt. of the pipe.



$$\sum F_x = 0 \Rightarrow N_A = N_B$$

$$\sum F_y = 0 \Rightarrow N_A \sin \theta/2 + N_B \sin \theta/2 = 0.3 N_A \cos \theta/2 + 0.3 N_B \cos \theta/2$$

$$2 N_A \sin \theta/2 = 0.6 N_A \cos \theta/2$$

$$(2 \sin \theta/2 - 0.6 \cos \theta/2) N_A = 0$$

$$2 \sin \theta/2 = 0.6 \cos \theta/2$$

$$\tan \theta/2 = 0.3$$

$$\tan \theta/2 = 0.3$$

$$\theta/2 = \tan^{-1}(0.3)$$

$$\theta = 33.4^\circ$$

• When the machine is not be able to move in reverse direction after the removal of load is called the self-locking machine.

When the machine is be able to move in reverse direction after the removal of load is called reversing machine.

The mean diameter of a screw jack is 60 mm. Find the effort required which will be applied at the end of the lever arm having length 900 mm.

- condition of self-locking Machine:
 where $\alpha \rightarrow$ helix angle
 $\phi \rightarrow$ angle of friction

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Find the following:-

Effort required to lift or raise a load of 30kN.

Effort required to lower the load of 30kN.

The efficiency for raising the load.

The efficiency for lowering the load.

Verify whether the screw Jack is self-locking or reversible.

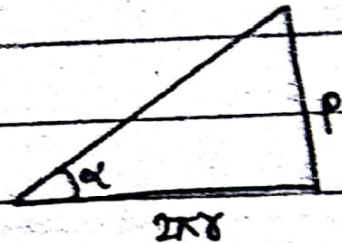
If the coefficient of friction of screw Jack is 0.08 and pitch of the screw is 12mm.

Pitch, $p = 12\text{mm}$

$$\alpha = \tan^{-1} \left(\frac{p}{\pi(30)} \right)$$

$$\alpha = \tan^{-1} \left(\frac{12}{60 \times 3.14} \right)$$

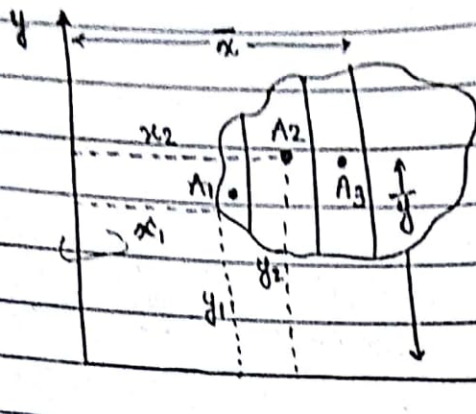
$$\alpha = 3.6^\circ$$



$\alpha \rightarrow$ helix angle



Centroid of an Area



$$A_1x_1 + A_2x_2 + A_3x_3 + \dots = A\bar{x}$$

$$\bar{x} = \frac{A_1x_1 + A_2x_2 + \dots}{A_1 + A_2 + \dots}$$

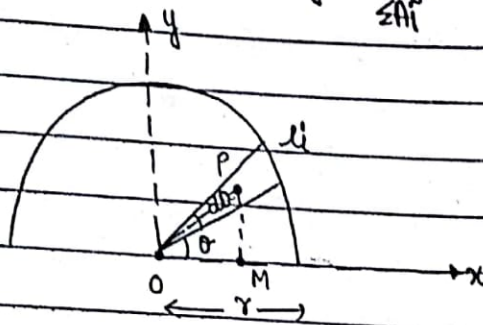
$$= \frac{\sum A_i x_i}{\sum A_i}$$

$$A_1y_1 + A_2y_2 + A_3y_3 + \dots = A\bar{y}$$

$$\bar{y} = \frac{A_1y_1 + A_2y_2 + \dots}{A_1 + A_2 + \dots}$$

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i}$$

Q.



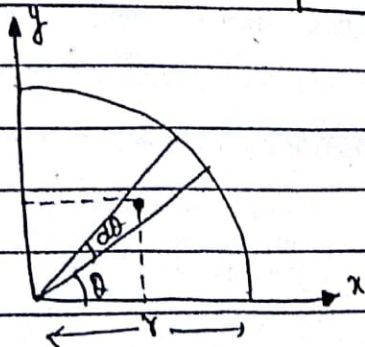
$$dA = \frac{1}{2}(bh) = \frac{1}{2}(r \sin \theta) r = \frac{1}{2} r^2 \sin \theta$$

$$OP = \frac{2}{3} r$$

$$\sin \theta = \frac{PM}{OP} \Rightarrow PM = \frac{2}{3} r \sin \theta = y_i$$

$$\bar{y} = \frac{\sum dA \times y_i}{\sum dA} = \frac{\int_0^{\pi/2} \frac{1}{2} r^2 \sin \theta \times \frac{2}{3} r \sin \theta}{\int_0^{\pi/2} \frac{1}{2} r^2 \sin \theta} = \frac{r^3 \int_0^{\pi/2} \sin^2 \theta \, d\theta}{\frac{r^2}{2} \int_0^{\pi/2} \sin \theta \, d\theta}$$

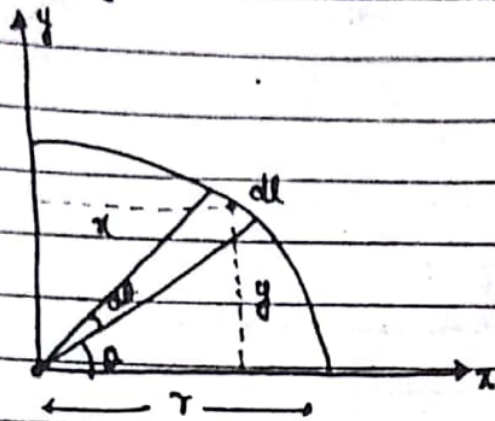
$$\bar{y} = \frac{2r}{3} \times \frac{2}{\pi} \Rightarrow \boxed{\frac{4r}{3\pi} = \bar{y}}$$



$$\bar{x} = \frac{4r}{3\pi}$$

$$\bar{y} = \frac{4r}{3\pi}$$

Taking line Element :



$$dl = r d\theta$$

$$\bar{y} = \frac{\int y dl}{\int dl}$$

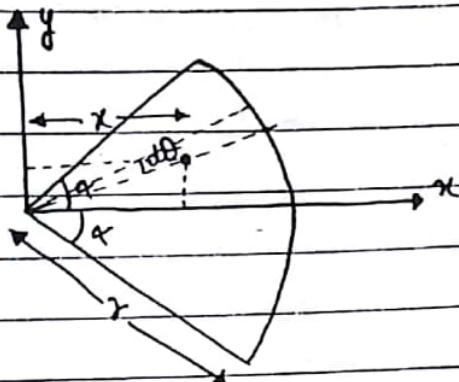
$$y = r \sin \theta$$

$$\bar{y} = \frac{\int r \sin \theta \cdot r d\theta}{\int r d\theta}$$

$$\bar{y} = \frac{\int_0^{\pi/2} r^2 \sin \theta d\theta}{\int_0^{\pi/2} r d\theta} = \frac{2r}{\pi}$$

$$\bar{x} = r \frac{\int_0^{\pi/2} \cos \theta d\theta}{\int_0^{\pi/2} d\theta} = \frac{2r}{\pi}$$

$$\boxed{\bar{y} = \frac{2r}{\pi}} \quad \text{and} \quad \boxed{\bar{x} = \frac{2r}{\pi}}$$



$$dA = \frac{1}{2} r^2 d\theta$$

$$\bar{x} = \frac{\int x dA}{\int dA} = \frac{\frac{2}{3} r^3 \int_{-\alpha}^{+\alpha} \cos \theta d\theta}{\int_{-\alpha}^{+\alpha} \frac{1}{2} r^2 d\theta}$$

$$\bar{x} = \frac{2r}{3} \cos \theta$$

$$\bar{x} = \frac{2r}{3} \left(\frac{2 \sin \alpha}{2\alpha} \right) = \frac{2r}{3\alpha} \sin \alpha = \bar{x}$$

$$y = \frac{2r}{3} \sin \theta, \quad \bar{y} = \frac{\int y dA}{\int dA} = \frac{\int_{-\alpha}^{+\alpha} \frac{r^2}{2} d\theta \cdot \frac{2r}{3} \sin \theta}{\int_{-\alpha}^{+\alpha} \frac{r^2}{2} d\theta}$$

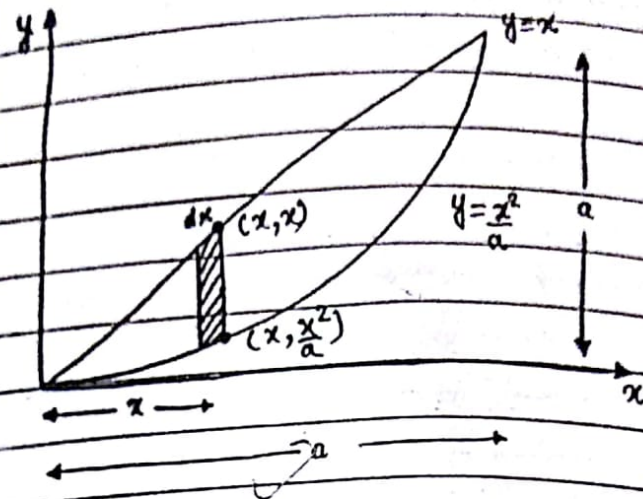
$$\bar{y} = \frac{2r}{3} \frac{\int_{-\alpha}^{+\alpha} \sin \theta d\theta}{\int_{-\alpha}^{+\alpha} d\theta} = \frac{2r}{3} \frac{(\cos \alpha - \cos \alpha)}{2\alpha} = 0$$

$$\boxed{\bar{y} = 0}$$

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find the centroid.



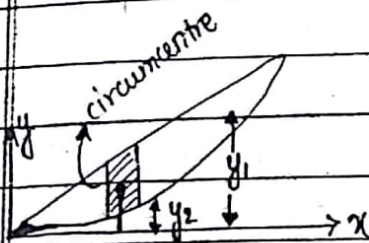
consider a small strip of thickness 'dx':

Area of this rectangular strip $\Rightarrow dA = (x - \frac{x^2}{a}) dx$

$$\bar{x} = \frac{\int_0^a x(x - \frac{x^2}{a}) dx}{\int_0^a (x - \frac{x^2}{a}) dx}$$

Area of strip
(Moment about y-axis)
Total area

$$\bar{y} = \frac{\int_0^a \frac{(y_1 + y_2)}{2} (x - \frac{x^2}{a}) dx}{\int_0^a (x - \frac{x^2}{a}) dx} = \frac{\int_0^a (x^2 - \frac{x^4}{a^2}) dx}{2 \int_0^a (x - \frac{x^2}{a}) dx}$$



$$\bar{x} = \int_0^a (x^2 - \frac{x^3}{a}) dx = \left[\frac{x^3}{3} - \frac{x^4}{4a} \right]_0^a$$

$$I \rightarrow \frac{y_1 + y_2}{2}$$

$$\left[\frac{x^2}{2} - \frac{x^3}{3a} \right]_0^a = \frac{a^2}{2} - \frac{a^2}{3}$$

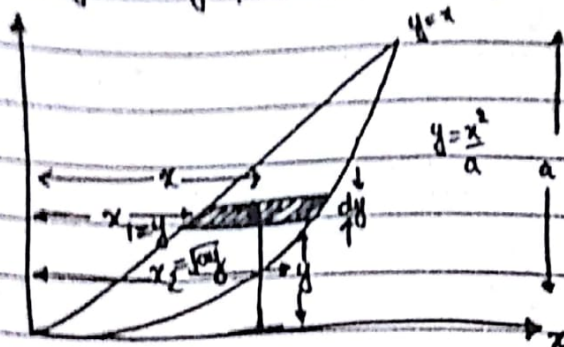
$$\text{height} \Rightarrow \frac{y_1 + y_2}{2} = \frac{x + \frac{x^2}{a}}{2}$$

$$\bar{x} \Rightarrow \frac{a^3}{3} - \frac{a^3}{4} \Rightarrow \frac{a^3}{12} = \frac{a^2}{12} \cdot \frac{a}{1} = \frac{a}{12}$$

$$\bar{y} = \frac{\left[\frac{x^3}{3} - \frac{x^5}{5a^2} \right]_0^a}{\frac{a^2}{2} - \frac{a^2}{3}} = \frac{\frac{a^3}{3} - \frac{a^3}{5}}{\frac{a^2}{6}} = \frac{\frac{2a^3}{15}}{\frac{a^2}{6}} = \frac{2a}{5}$$

OR

By taking horizontal strip:-



$$dA = (x_2 - x_1)dy$$

$$dA = (\sqrt{ay} - y)dy$$

$$x = \frac{x_1 + x_2}{2} = \frac{y + \sqrt{ay}}{2}$$

$$\bar{x} = \frac{\int dA \cdot x}{\int dA}$$

$$\bar{y} = \frac{\int_0^a y \cdot (\sqrt{ay} - y) dy}{\int_0^a (\sqrt{ay} - y) dy} = \left[\frac{\sqrt{a} \cdot y^{3/2}}{3/2} - \frac{y^3}{3} \right]_0^a$$

$$\int_0^a (\sqrt{ay} - y) dy$$

$$\left[\frac{\sqrt{a} y^{3/2}}{3/2} - \frac{y^3}{3} \right]_0^a$$

$$= \left[\frac{2\sqrt{a}}{5} y^{5/2} - \frac{y^3}{3} \right]_0^a$$

$$= \frac{2}{5} a^3 - \frac{a^3}{3} = \frac{a^3}{15}$$

$$\frac{2a^2}{3} - \frac{a^2}{2}$$

$$\frac{a^2}{6}$$

$$\frac{a^2}{6} = \frac{2a}{5}$$

$$\bar{x} = \frac{\int_0^a \left(\frac{y + \sqrt{ay}}{2} \right) (\sqrt{ay} - y) dy}{\int_0^a (\sqrt{ay} - y) dy} = \frac{\int_0^a \left(\frac{ay}{2} - \frac{\sqrt{a} y^{3/2}}{2} \right) dy}{\int_0^a (\sqrt{ay} - y) dy} = \left[\frac{ay^2}{2} - \frac{2\sqrt{a} y^{5/2}}{5} \right]_0^a$$

$$\frac{a^2}{6}$$

$$\frac{a^2}{6}$$

$$\frac{a^2}{6}$$

$$\bar{x} = \frac{\int_0^a (ay - y^2) dy}{\int_0^a (\sqrt{ay} - y) dy} = \left[\frac{ay^2}{2} - \frac{y^3}{3} \right]_0^a = \frac{a^3}{2} - \frac{a^3}{3} = \frac{a}{2}$$

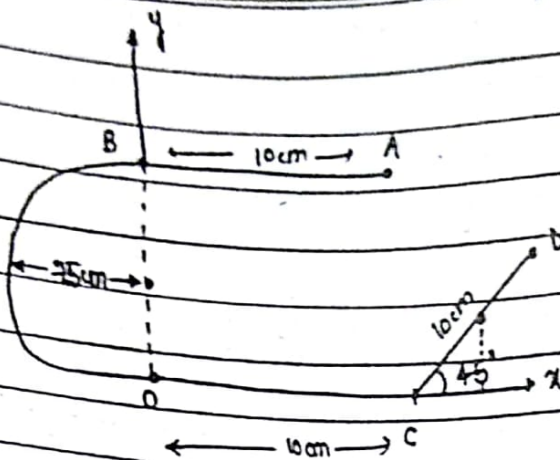
$$2 \cdot \frac{a^2}{6}$$

$$\frac{a^2}{3}$$

$$\frac{a^2}{3}$$

$$\frac{a}{2}$$

Q.



$$l_1 = 10 \text{ cm}$$

$$x_1 = 5 \text{ cm}$$

$$l_2 = 10 \text{ cm}$$

$$x_2 = 5 \text{ cm}$$

$$\bar{x} = \frac{\sum l_i x_i}{\sum l_i}$$

$$l_3 = \pi \times 7.5 \text{ cm}$$

$$x_3 = -2(7.5) \text{ cm}$$

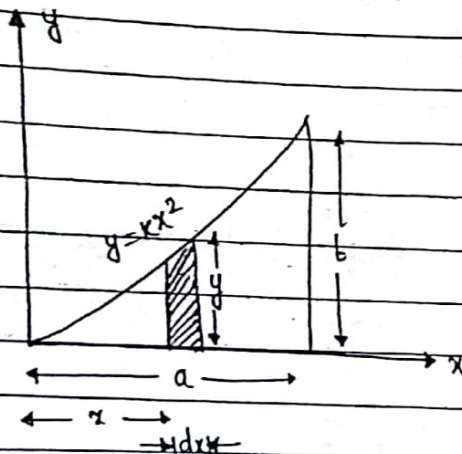
$$\bar{x} = \frac{\sum l_i x_i}{\sum l_i} = \frac{l_1 x_1 + l_2 x_2 + l_3 x_3}{l_1 + l_2 + l_3 + l_4}$$

$$l_4 = 10 \text{ cm}, x_4 = 10 + \frac{5}{\sqrt{2}}$$

or is the reference axis and all the distances are measured from the axis.

$$\bar{y} = \frac{\sum l_i y_i}{\sum l_i} = \frac{l_1 y_1 + l_2 y_2 + l_3 y_3 + l_4 y_4}{l_1 + l_2 + l_3 + l_4}$$

or is the reference axis.



$$dA = y dx$$

$$\bar{y} = \frac{\int dA \cdot y}{\int dA} \quad \text{and} \quad \bar{x} = \frac{\int dA \cdot x}{\int dA}$$

$$\bar{y} = \frac{1}{2} \int kx^2 \cdot dx \cdot kx^2$$

$$\int kx^2 dx$$

$$\bar{y} = \frac{k \int_0^a x^4 dx}{\int_0^a x^2 dx} = \frac{3ka^3}{10}$$

$$\text{when } x=a; y=b \Rightarrow b=ka^2 \Rightarrow$$

$$\therefore \bar{y} = \frac{3ab}{10}$$

$$\boxed{\bar{x} = \frac{b}{a^2}}$$

α → helix angle.

condition of self-locking

machine : $\alpha < \phi$

helix angle

angle of friction

2) η (efficiency) $< 50\%$

$$\alpha = \tan^{-1} \left(\frac{p}{2\pi (30)} \right)$$

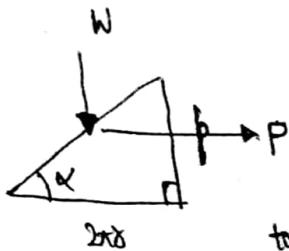
$$\alpha = \tan^{-1} \left(\frac{12}{60 \times 3.14} \right)$$

$$\alpha = 3.64^\circ$$

$$\mu = 0.08$$

$$\phi = \tan^{-1}(\mu) = 4.57^\circ \text{ (Angle of friction)}$$

$\phi > \alpha \Rightarrow$ So our screw jack is a self-locking machine.



$$P_H^+ = W \tan(\phi + \alpha) = 30000 \times \tan(4.57 + 3.64)$$

$$\text{Minimum effort required} = 4328 \text{ N}$$

to raise the load in upward dir.ⁿ in screw jack

$$P_{\text{lever}}^+ = P_H^+ \times \frac{r}{l} = \frac{4328 \times 30}{900} = 144.26 \text{ N}$$

P_H^- = Minimum effort required to lower the load at

$$\text{the mean radius} = W \tan(\phi - \alpha) = 30000 \tan(4.57 - 3.64)$$

$$= 486.9 \text{ N}$$

Minimum effort required to lower the load at the lever:

$$P_L^- = P_H^- \times \frac{r}{l} = 16.23$$

Efficiency of raising the load $\Rightarrow \eta_+ = \frac{P_{\text{ideal}}}{P_{\text{actual}}} \times 100$

P_{ideal} which is applied at mean radius

is given by, $W \tan(\alpha) = P_{\text{ideal}}$

$$P_{\text{ideal}} = 1908.46 \text{ N}$$

$$\Rightarrow \frac{1908}{4328} \times 100$$

$$\Rightarrow 44\%$$

pressure →

velocity →

stress →

Equilibrium of forces →

Efficiency for lowering the load $\Rightarrow \eta_- = \frac{486.9 \times 100}{1908} \Rightarrow 25.5\%$